

AI-Assisted Quality Gates: A Multi-Stage Validation Workflow



Original Input

Problem & Answer Pair

- Raw Data from Banks
- Diverse Problem Sources
- Baseline for Refinement
- Initial Material



Cleaner

Candidate Generation

- Apply Domain Conventions
- Use Repair Principles
- Generate Formal Candidates
- Correct Problem Pairs



Reviewer

Consistency Check

- Logical Consistency Check
- Factual Alignment Guard
- Block Unsupported Changes
- Gatekeeper



Validation

Automated Test

- Test Gurobi Code
- Solution Feasibility
- Numerical Stability
- Optimization Constraints

✗ Intercepted

Rejected due to factual misalignment, logical contradictions, or unsupported arbitrary edits to the original problem.

⚠ Failed

Triggers the Diagnostic Resolver for root-cause analysis. Results in either auto-repair or escalation for manual expert review.

Total Filtered: 1,119 Samples

✓ Approved

Consistency verified. The candidate moves forward to the critical automated code execution stage for functional testing.

✓ Passed

Sample satisfies all functional and quality benchmarks. Proceeds to the final curated dataset for model training.

Final Dataset: 8,881 Validated Samples